

**Constraining Solid Dynamics, Interface Rheology, and Slab Hydration
in the Hikurangi Subduction Zone Using 3D Fully Dynamic Models**

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Introduction

This supplement document provides additional information regarding the numerical methods and data post processing for “Constraining Solid Dynamics, Interface Rheology, and Slab Hydration in the Hikurangi Subduction Zone Using 3D Fully Dynamic Models”. We first outline the approximation for the Peierls creep deformation mechanism used in this study, then we discuss our velocity misfit calculation in more detail and present our calculation for modeled trench retreat.

Peierls Creep Approximation

We use an approximate version the Peierls creep mechanism (equation 5) for computational efficiency. This approximate version is expressed by:

$$\begin{aligned}\eta_p &= A \cdot B \cdot C \\ A &= \frac{0.5\gamma\sigma_p}{(A_p(\gamma\sigma_p)^n)^{\frac{1}{s+n}}} \\ B &= \exp\left(\frac{E+PV}{RT} \cdot p \cdot q \cdot (1-\gamma^p)^{q-1} \cdot \gamma^p\right) \\ C &= \dot{\epsilon}^{\frac{1}{s+n}-1} \\ s &= \frac{pq(E+PV)}{RT} \cdot (1-\gamma^p)^{q-1} \gamma^p\end{aligned}$$

where γ is a fitting parameter set to 0.17 and other parameters are defined in Table 1 in the main publication. A value of 0.17 is chosen for γ because this is the most appropriate value for the high stresses within the subducting plate.

Calculating Velocity Misfit

We determine the misfit in convergence rate between the ASPECT models (Figure S1a) and the MORVEL velocities (Figure S1b) at every point on the modeled Pacific plate east of the slab. Therefore, we do not calculate the misfit in convergence north of 26°S where we terminate the slab for computational efficiency (Figure S1c). The largest misfit occurs close to the trench (Figure S1c), where the modeled convergence transitions from dominantly east-west convergence to include a radial (downward) component. The misfit peaks at the northern extent of the modeled slab.

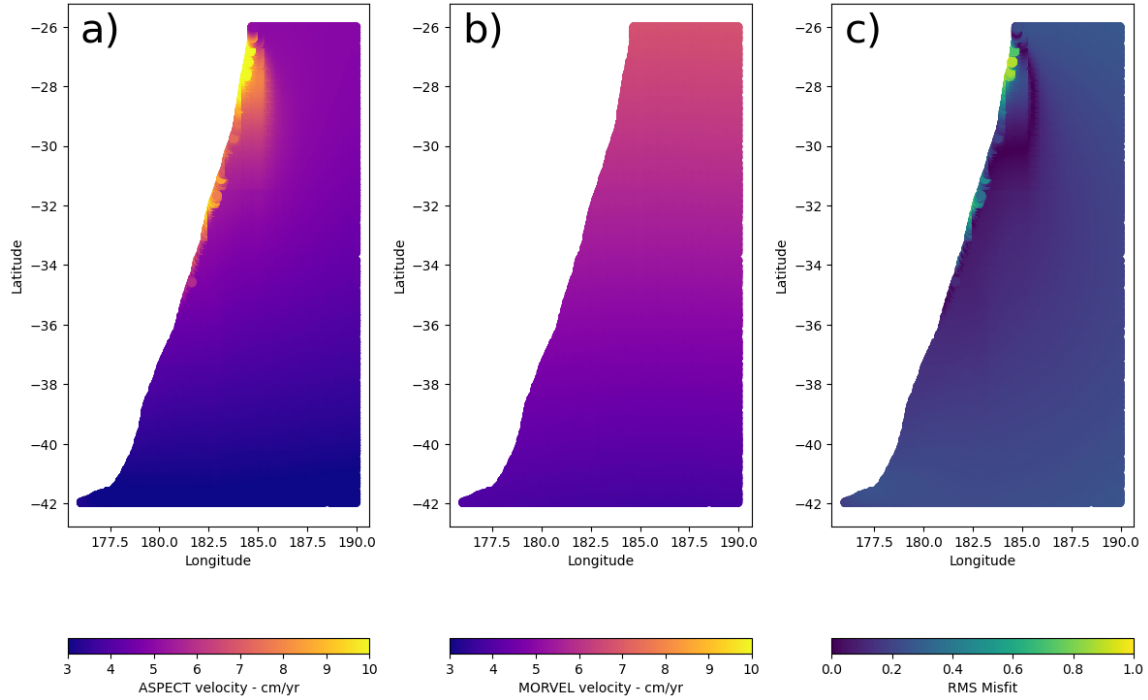


Figure S1. Surface velocities of the Pacific plate from (a) the best fitting ASPECT model, and (b) the MORVEL model (DeMets et al., 2010). The RMS misfit between the ASPECT velocities and the MORVEL velocities is shown in panel c).

Modeled Trench Motion

To calculate the migration of the trench, we extract the location of the modeled trench and draw a parallel curve on either side. Next, we query the magnitude of the velocity at each point on both of these trench-parallel curves and determine the trench motion using

$$v_{trench} = v_{Pac} - v_{Aus},$$

where, v_{trench} , v_{Pac} , and v_{Aus} is the velocity of the trench, the velocity of the Pacific plate, and the velocity of the Australian plate, respectively. Therefore, when the velocity is negative, the trench retreats, and when the velocity is positive, the trench advances.

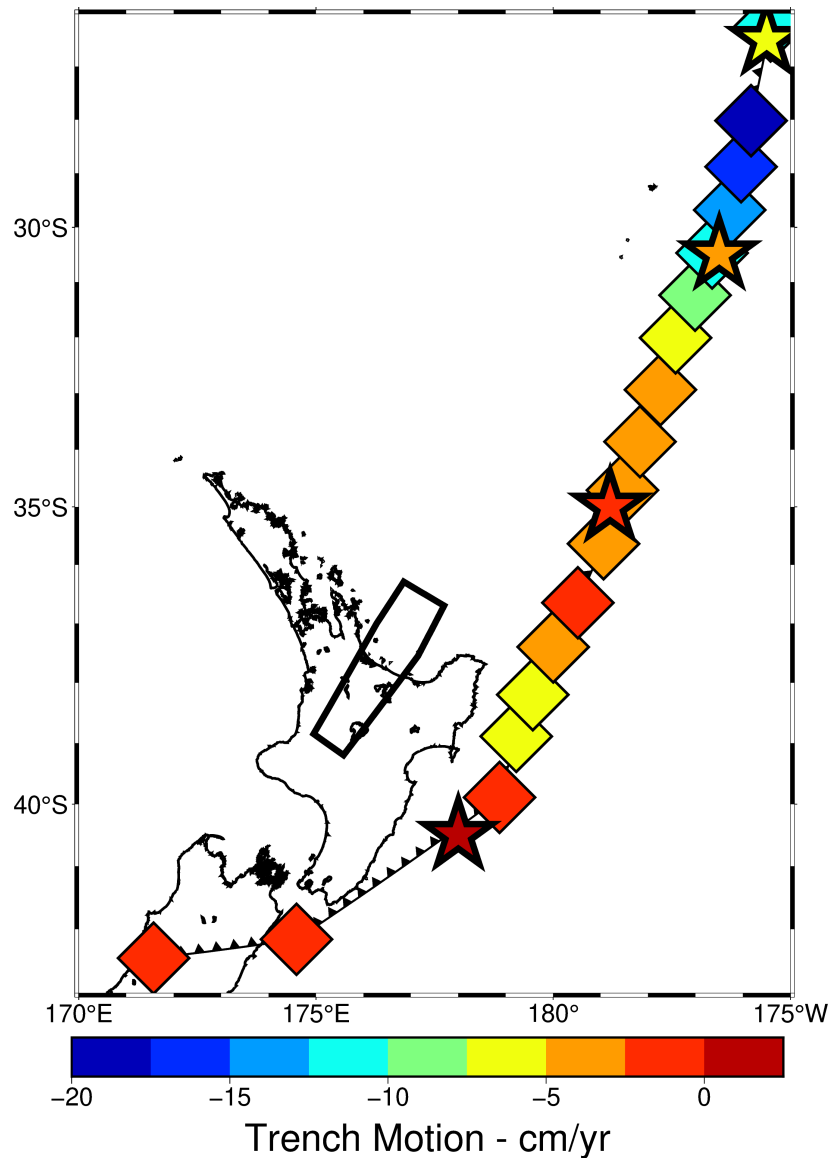


Figure S2. The modeled trench motion (coloured diamonds) at each point of the modeled trench as defined by the Geodynamic WorldBuilder compared to estimates of trench motion from Schellart and Spaceman, 2012 (coloured stars). The black sawtooth line shows the location of the modeled trench, and the solid black contour shows the coastline of New Zealand. Negative (positive) trench motion corresponds to trench retreat (advance)

Bibliography

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